

Ex. No: 02

Image Capture and Video Processing Using OpenCV

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In [59]: import cv2
import matplotlib.pyplot as plt
from IPython.display import clear_output
import time
```

```
In [44]: cap = cv2.VideoCapture(0)

ret, frame = cap.read()

if ret:
    cv2.imwrite("captured_image.jpg", frame)

cap.release()
```

```
In [45]: captured_image=cv2.imread("captured_image.jpg")
```

```
In [46]: plt.imshow(captured_image[:, :, ::-1])
plt.title('Captured Frame')
plt.axis('off')
plt.show()
```

Captured Frame



```
In [47]: cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()

    if not ret:
        break

    frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    clear_output(wait=True)
    plt.imshow(frame)
    plt.axis("off")
    plt.show()

    time.sleep(0.05)

cap.release()
```



In []:

```
In [ ]: cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()

    if not ret:
        break

    # Resize the frame
    resized_frame = cv2.resize(frame, (100, 150))
    # Convert BGR to RGB
    resized_frame = cv2.cvtColor(resized_frame, cv2.COLOR_BGR2RGB)

    clear_output(wait=True)
```

```
plt.imshow(resized_frame)
plt.axis("off")
plt.show()

time.sleep(0.05)

cap.release()
```



```
In [49]: cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()

    if not ret:
        break

    # Rotate the frame
    rotated_frame = cv2.rotate(frame, cv2.ROTATE_90_CLOCKWISE)

    # Convert BGR to RGB
    rotated_frame = cv2.cvtColor(rotated_frame, cv2.COLOR_BGR2RGB)

    clear_output(wait=True)
    plt.imshow(rotated_frame)
    plt.axis("off")
    plt.show()

    time.sleep(0.05)

cap.release()
```



In []:

In []:

In []: